

Objectives:

1. **Literature Review:** Read and summarise research papers relevant to the program. Extract for each: a) the clinical or engineering problem (Q1), b) the method used (Q2), c) the reported outcome (Q3), d) the authors' own stated limitation (Q4). Write 3–5 sentences per paper, using the three-pass research method.
2. **Gap Analysis:** Name two gaps you could plausibly target with a 12-week pilot. A gap is not “no one has done this yet” — it is “this was tried and here is what was missed.”
3. **Preliminary Proposal:** Complete using the template (90 words or less), including the problem statement, proposed solution, summary of previous work and references.

Task 1: Literature Review

- Tyler, S., Olis, M., Aust, N., Patel, L., Simon, L., Triantafyllidis, C., Patel, V., Lee, D. W., Ginsberg, B., Ahmad, H., & Jacobs, R. J. (2024). *Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review*. *Cureus*, 16(5), e59906. <https://doi.org/10.7759/cureus.59906>

Paper 1 – This scoping review examines how traditional emergency room triage (where staff manually sort patients into five urgency levels) often suffers from human error, high subjectivity, and emergency department overcrowding. To investigate this, the authors reviewed 29 primary research studies (in 2023) published over a 10 year period in major medical databases. The review found that many machine learning models achieved higher accuracy than traditional triage methods and were able to make predictions more quickly. However, a major downside noted by the authors is that these AI tools often assign patients to higher urgency categories than necessary, which can accidentally waste valuable hospital resources, and the models lack a clear way to explain why they made a decision which may make clinical adoption more difficult.

- Da’Costa, A., Teke, J., Origbo, J. E., Osonuga, A., Egbon, E., & Olawade, D. B. (2025). *AI-driven triage in emergency departments: A review of benefits, challenges, and future directions*. *International Journal of Medical Informatics*, 197, 105838. <https://doi.org/10.1016/j.ijmedinf.2025.105838>

Paper 2 – This paper focuses on the clinical problem of emergency room overcrowding, uneven patient sorting during peak hours, and the heavy cognitive stress placed on hospital staff. The authors conducted a narrative review of peer-reviewed articles published over a 9 year period across databases like PubMed and Google Scholar. The reported outcomes show that AI tools were associated with shorter wait times, more consistent triage decisions and improved resource allocation (hospital beds, staffing). However, a major limitation the authors mentioned is that these systems can carry algorithmic biases that treat diverse patient populations unfairly, and there is currently an uncertainty around responsibility when AI systems make mistakes.

- Araouchi, Z., & Adda, M. (2024). *TriageIntelli: AI-Assisted multimodal triage system for health centers*. *Procedia Computer Science*, 251, 430–437. <https://doi.org/10.1016/j.procs.2024.11.130>

Paper 3 – The authors investigated emergency room overcrowding and human judgement errors under the structured Korean Triage and Acuity Scale (KTAS). The authors trained six machine learning models on 1,267 emergency department records and used data-balancing techniques to address unequal class sizes. The reported outcomes show that an integrated “Stacking Model” achieved the highest classification accuracy of 80.05%, but the individual models had accuracies of roughly 78–80% standalone. A limitation identified by the authors is that the system relies on a single dataset source, which creates a need for further testing in other hospitals to ensure it functions reliably across different hospital networks.

- Bhuvaneswari, E., Prasad, K. D. V., Ashraf, M., Jadhav, S., Rao, T. R. K., & Rani, T. S. (2025). *A human-centered hybrid AI framework for optimizing emergency triage in resource-constrained settings*. *Intelligence-Based Medicine*, 12, 100311. <https://doi.org/10.1016/j.ibmed.2025.100311>

Paper 4 – This study explores the issue of high human error rates, cognitive burnout and misallocated medical supplies in overcrowded emergency rooms operating under severe resource constraints. To solve this, the authors developed a hybrid AI prototype combining a machine learning model for risk prediction, an explanation feature showing why the model made a recommendation, and a resource management component that considered hospital bed and oxygen availability. The reported outcomes show that the system achieved a high 90% accuracy rate, reduced clinician decision-making time by 31% and improved the allocation of hospital resources by 20%. However, a key limitation stated by the authors is that the system was developed using a very small sample of real patient data, meaning it still requires testing in larger and more diverse hospital settings to determine whether it performs reliably.

- Huang, J., Galal, G., Etemadi, M., & Vaidyanathan, M. (2022). *Evaluation and Mitigation of Racial Bias in Clinical Machine Learning Models: Scoping Review*. *JMIR Medical Informatics*, 10(5), e36388. <https://doi.org/10.2196/36388>

Paper 5 – This review examines racial bias in machine learning models, which can worsen existing healthcare inequalities if not addressed. To analyse this issue, the authors performed a systematic scoping review following standard PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-analyses) guidelines to screen 635 research papers, ultimately analysing 12 core studies that evaluated algorithmic fairness. The reported outcomes revealed that while 67% of clinical models showed evidence of racial bias, all reviewed studies that actively applied technical corrections successfully improved model fairness. However, a major limitation highlighted by the authors is that medical algorithms suffer from a severe lack of standardised reporting and missing demographic data, making it difficult for hospitals to reliably evaluate whether a tool is safe and unbiased before deploying it.

- Agius, S., Cassar, V., Magri, C., Khan, W., Obe, D. A.-J., Caruana, G., & Topham, L. (2025). *Predicting Emergency Severity Index (ESI) level, hospital admission, and admitting ward in an emergency department using data-driven machine learning*. BMC Medical Informatics and Decision Making, 25, 281. <https://doi.org/10.1186/s12911-025-02941-9>

Paper 6 – This study looks at emergency department bottlenecks, unexpected patient surges, and the difficulty of accurately predicting hospital resource and bed needs at initial triage. To resolve this, the authors used the XGBoost machine learning algorithm to develop a two-stage model trained on a dataset of over 650,000 patient visits across 6 years. The model accurately predicted triage urgency, likelihood of admission and admitting ward, with performance improving when blood test results were included. However, a major stated limitation is that the system was developed using data from a single national hospital network, which creates a gap for external testing to ensure it works reliably in different regional or international health systems.

- Almulihi, Q. A. A., Alquraini, A. A., Almulihi, F. A. A., Alzahid, A. A., Al Jathnan Al Qahtani, S. S., Almulhim, M., Alghtani, S. H. S., Alnafea, F. M. N., Mushni, S. A. S., Alaqil, N. A., Assiri, M. I. F., & Maghraby, N. H. (2024). *Applications of Artificial Intelligence and Machine Learning in Emergency Medicine Triage – A Systematic Review*. Medical Archives, 78(3), 198-206. <https://doi.org/10.5455/medarh.2024.78.198-206>

Paper 7 – This systematic review evaluates severe emergency department overcrowding and the high potential for human error when clinicians attempt to sort complex, high-risk patient data under extreme time pressure. To investigate this, the authors executed a systematic review following standard PRISMA guidelines to filter and analyse recent peer-reviewed literature detailing machine learning tools in emergency triage. The review found that AI systems can improve the speed and accuracy of triage decisions when identifying high-risk patients. However, a major limitation stated by the authors is that most existing models suffer from limited dataset diversity and a lack of real-world clinical testing, which limits confidence in how well these systems would perform in different hospital settings.

- Kim, J., Shin, J., Kim, K., & Chang, H. (2024). *Deep learning-based natural language processing for detecting medical symptoms and histories in emergency patient triage*. American Journal of Emergency Medicine, 76, 203-210. <https://doi.org/10.1016/j.ajem.2023.11.063>

Paper 8 – This study investigates the issue of unused data in emergency departments where important clinical information is often recorded in free-text nursing notes that are difficult for traditional systems to analyse. The authors evaluated several natural language processing models designed to extract symptoms and medical history from triage notes. The best performing models achieved F1-scores above 88%, outperforming standard keyword based approaches. However, a major limitation stated by the authors is that the algorithm's performance depends heavily on the specific writing style and shorthand abbreviations of a single hospital, meaning it requires localised adjustment and retraining to deal with typographical errors or different medical slang elsewhere.

- Tschoellitsch, T., Seidl, P., Böck, C., Maletzky, A., Moser, P., Thumfart, S., Giretzlehner, M., Hochreiter, S., & Meier, J. (2023). *Using emergency department triage for machine learning-based admission and mortality prediction*. European Journal of Emergency Medicine, 30(6), 408–414. <https://doi.org/10.1097/MEJ.0000000000001064>

Paper 9 – The authors examined the clinical problem of inefficient patient flow and unpredictable bed demands caused by traditional triage frameworks, which only measure immediate urgency rather than predicting downstream hospital admissions or patient mortality. To solve this, the authors used Random Forests and Neural Networks to analyse a dataset of 58,323 emergency patient records collected within ten minutes of arrival at a major Austrian hospital. The reported outcomes demonstrate that the machine learning models achieved high accuracy in predicting ward admissions and ICU placements, with age and chief complaint being important predictors. However, a key limitation identified by the authors is that the study was conducted in a single hospital facility and did not include physiological vital signs or blood test results, meaning the model requires further expansion and multi-centre testing to verify its safety elsewhere.

Task 2: Gap Analysis

Based on the literature review, the following gaps were identified. Gap 1 is selected as the primary focus for the proposed 12-week project.

Gap 1 (Primary Project Focus): Explainable AI + Clinician Trust

Evidence: Tyler et al. (2024) found that many AI triage models cannot explain the reason behind their recommendations. Da’Costa et al. (2025) identified clinician trust as a major barrier to adoption and emphasised the need for transparent systems. Bhuvaneswari et al. (2025) proposed a human-centred framework that incorporates explainable AI techniques, highlighting that accuracy alone may not be enough for clinical adoption. Kim et al. (2024) further showed that AI models can extract useful clinical information from nursing notes but their outputs remain difficult to verify without clear explanations.

What was missed: Many studies optimise for predictive accuracy, yet comparatively fewer focus on making recommendations understandable and verifiable to clinicians.

Pilot Idea: Develop an AI triage assistant that predicts a triage level and displays the factors influencing each recommendation.

Gap 2: Human-AI Collaboration

Evidence: Da’Costa et al. (2025) emphasised the importance of clinician trust, education and human oversight in AI-assisted triage. Bhuvaneswari et al. (2025) proposed a human-centred triage framework in which AI supports rather than replaces clinical decision-making. Almulihi et al. (2024) found that although AI systems often outperform traditional approaches, most studies lack real-world clinical deployment and evaluation. Together, these findings suggest that successful implementation depends not only on model performance but also on how clinicians interact with AI recommendations.

What was missed: Most studies evaluate whether AI performs better than clinicians, rather than whether clinicians perform better when supported by AI.

Pilot Idea: Evaluate whether AI decision support improves consistency among junior triage nurses when assigning patient acuity levels.

Gap 3 (Future Work): Performance Across Different Hospitals

Evidence: Araouchi & Adda (2024) reported that their highest performing triage model was developed using data from a single healthcare institution and explicitly identified the need for multi-centre validation. Da'Costa et al. (2025) highlighted concerns regarding algorithmic fairness and the need for testing across diverse patient populations. Agius et al. (2025) developed machine learning models for predicting Emergency Severity Index (ESI) level and hospital admissions but noted that validation was limited to a single emergency department. Similarly, Tschoellitsch et al. (2023) demonstrated strong performance in admission and mortality prediction using emergency triage data, yet their study was conducted in one Austrian hospital and lacked external validation. Kim et al. (2024) also found that natural language processing (NLP) models depend heavily on local documentation styles, suggesting that performance may vary when applied in different healthcare settings.

What was missed: Many AI triage models are developed and tested using data from a single hospital, healthcare network or regional population. As a result, it remains unclear whether these models would perform as well in different healthcare settings.

Pilot Idea: Evaluate an existing triage model using a secondary dataset or simulated cross-site validation. This raises an important question: “Does an AI model trained in one setting remain reliable in another?”

Gap 1: Can clinicians understand the AI?

Gap 2: Can clinicians work effectively with the AI?

Gap 3: Does the AI still work elsewhere?

Literature Synthesis (Summary of the 9 Papers)

Across the reviewed literature, there is broad agreement that artificial intelligence can improve the speed, consistency and accuracy of emergency department triage. Multiple studies reported successful prediction of triage levels, hospital admissions, mortality risk and resource requirements using machine learning techniques.

However, the literature also brings up recurring challenges. Many models operate as black boxes that provide little explanation for their recommendations, making them difficult for clinicians to understand and verify. Most systems have been developed and validated using data from a single institution, raising concerns about generalisability and fairness across different patient populations. Together, these studies suggest that future work should focus on making AI recommendations easier for clinicians to understand and trust, while ensuring that models remain reliable across different healthcare settings.

Task 3: Preliminary Proposal

CariSurg MedTech Pathways Research Project

Preliminary Proposal

Project Title: Explainable AI Triage Assistant for Caribbean Emergency Departments

Student Name: Sariana Ramoutar

Statement of the Problem:

Emergency departments rely on rapid triage decisions to prioritise patient care, yet heavy workloads, patient surges and human subjectivity can contribute to inconsistent triage outcomes. Recent studies have shown that artificial intelligence can support emergency department triage by improving speed and predictive performance, but many systems operate as “black boxes,” providing little explanation for how recommendations are made (Tyler et al., 2024; Da’Costa et al., 2025). This lack of explainability can make AI recommendations difficult for clinicians to understand, verify and confidently incorporate into their decision-making processes, limiting their practical use in healthcare settings (Bhuvaneswari et al., 2025).

Previous Work:

Recent research demonstrates that artificial intelligence can improve the speed, consistency and accuracy of emergency department triage. Tyler et al. (2024), Da’Costa et al. (2025) and Almulihi et al. (2024) reported that machine learning systems frequently outperform traditional triage approaches and can support patient prioritisation, admission prediction and resource planning. Bhuvaneswari et al. (2025) highlighted the importance of human-centred AI in resource-constrained settings while Kim et al. (2024) demonstrated the potential of AI to extract clinically relevant information from triage notes. Huang et al. (2022) further emphasised the need to evaluate fairness and bias in clinical machine learning systems. Several studies [Araouchi and Adda (2024), Agius et al. (2025), Tschoellitsch et al. (2023)] also reported that models were typically developed and tested within a single institution, limiting evidence of performance in other healthcare settings. Collectively, these studies suggest that limited explainability, concerns about clinician trust and uncertainty about performance across healthcare settings remain important barriers to implementation.

Proposed Solution:

Develop and evaluate an AI decision-support prototype for emergency department triage using a de-identified dataset. In addition to predicting patient triage levels, the system will provide a clear explanation of the clinical factors contributing to each recommendation. The project will evaluate whether an explainable AI model can achieve triage classification performance comparable to published studies while displaying the factors influencing each prediction. The project will use publicly available emergency department triage data and is intended as an initial step toward explainable triage tools suitable for resource-constrained Caribbean healthcare settings.

References:

- Tyler, S., Olis, M., Aust, N., Patel, L., Simon, L., Triantafyllidis, C., Patel, V., Lee, D. W., Ginsberg, B., Ahmad, H., & Jacobs, R. J. (2024). *Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review*. *Cureus*, 16(5), e59906. <https://doi.org/10.7759/cureus.59906>
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